

The net sown area of West Bengal is approximately 60% while it is 40% in Andhra Pradesh. Thus, statement (a) is correct. The Laterite soil is formed in heavy rainfall areas by leaching. It is well developed in parts of Western Ghats, Eastern Ghats, South Maharashtra, parts of Karnataka, Andhra Pradesh, Odisha, West Bengal, Kerala, Jharkhand, Assam, and Meghalaya. Andhra Pradesh has Red soil and Alluvial soils in most parts of its area. Thus, reason (R) is not correct.

6. **Laterite soils is found in :**

- (a) Uttar Pradesh (b) Himachal Pradesh  
(c) Punjab (d) Maharashtra

**U.P.U.D.A./L.D.A. (Pre) 2006**

**Ans. (d)**

Laterite soil is found in typical monsoon conditions such as high temperature and heavy rainfall with wet and dry periods. The alterations of the wet and dry seasons leads to the leaching away of siliceous matter and lime from the rocks, leaving behind a soil rich in oxides of iron and aluminum compounds. These are leached soils of wet humid tropical regions, which are less fertile. They are well developed in parts of Western Ghats, Eastern Ghats, South Maharashtra, parts of Karnataka, Andhra Pradesh, Odisha, West Bengal, Kerala, Jharkhand, Assam and Meghalaya. Laterite soils are widely spread in India and cover an area of 2.48 lakh sq km. Laterite soil is mostly spread in Kerala (Malabar coastal region) followed by Maharashtra. Kerala is not present in the options so the most probable answer would be Maharashtra.

7. **Laterite soils are predominant in:**

- (a) Malabar Coastal Region  
(b) Coromandel Coastal Region  
(c) Bundelkhand  
(d) Baghelkhand

**U.P.P.C.S. (Pre) 2000**

**Ans. (a)**

See the explanation of the above question.

8. **Which one of the following statements is not true for Laterite soils:**

- (a) They are the soils of the humid tropical regions.  
(b) They are highly leached soils  
(c) Their fertility is low  
(d) They are rich in lime

**U.P.P.C.S.(Pre) 2001**

**Ans. (d)**

See the explanation of the above question.

9. **Which one of the following soil types of India is rendered infertile by the presence of excess iron?**

- (a) Desert sand (b) Alluvial  
(c) Podzolic (d) Lateritic

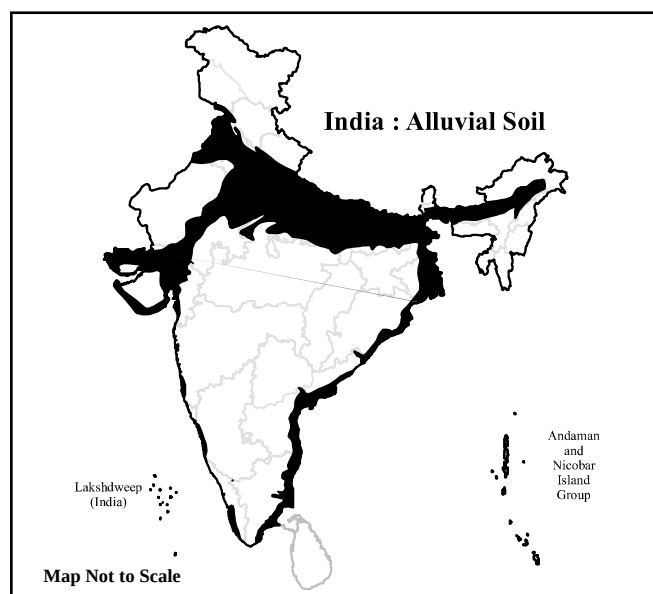
**I.A.S. (Pre) 1994**

**Ans. (d)**

Among the above soils, only Laterite soils have iron oxides. Hence, Laterite soil of India is rendered infertile by the presence of excess iron.

### iii. Alluvial Soil

\*Alluvial soils are widespread in the northern plains and the river valleys. These soils cover about 15 lakh sq. km or about **45.6%** of the total area of the country. \*The alluvial soil is largely found in the Indo-Gangetic plain starting from **Punjab** in the west to the West Bengal and Assam in the east. These soils are also found in deltas of Mahanadi, Godavari, Krishna and Cauvery, where they are called deltaic Alluvium. Alluvial soils are formed mainly by Himalayan sediments and sea retreats.



\*The colour of Alluvial soil varies from light grey to ash grey. They vary in nature from **Sandy, Loam to Clay**. \*Generally they are rich in **Phosphoric Acid, Potash, Lime, and Organic Matter** but poor in **Nitrogen and Humus content**. \*Due to the cultivation of leguminous crops nitrogen fixation takes place at a faster rate in these soils. They are suitable for irrigation which helps in growing crops like Wheat, Rice, Sugarcane, Jute, Cotton, Maize, Oilseed, Fruits and Vegetables.

\*Generally loamy soil has 40% sand particles, 18% clay

particles and 42% silt particles.

**\*Types of Alluvial Soil :**

**Khadar** - Khadar is the **new alluvium** and is deposited by floods annually enriching the soil by depositing fine silts. It always remains fertile.

**Bangar** - Bangar represents a system of **older alluvium** deposited away from the flood plains. Its color is pale reddish brown and in this, nodules are formed at large scale.

**Bhabar** - Bhabar is the **gently sloping coarse alluvial zone** below the Shiwalik Hills, where streams disappear into permeable sediments.

**Tarai** - The **marshy lowland region south of the Bhabar** is called Tarai (Terai). The soil of this region is generally rich in nitrogen and organic matter but poor in phosphate. The ancient alluvial soil found at higher altitudes in this region is known as **Rarh Plain**.

**1. Choose the correct statement out of the four statements given regarding alluvial soil :**

- (a) This soil is found in about 14% portion of India's land surface
- (b) Phosphoric acid is found in rich quantity in this soil
- (c) This soil is also known by names such as Reh, Thur, Chopan
- (d) This soil is not fertile

**Chhattisgarh P.C.S. (Pre) 2020**

**Ans. (b)**

Alluvial soils : They are the largest group, covering about 15 lakh sq. km or about 45.6% of the total area. Alluvial soils have Potash, Phosphoric acid, and Alkalies in sufficient quantity, but the proportion of nitrogen is low. Alluvial soils as a whole are very fertile.

Reh, Thur, Chopan, Kallar, Rakar, Karl etc. are the local names for saline soils.

**2. Which one of the following is the most productive soil in India?**

- (a) Red soil
- (b) Black soil
- (c) Alluvial soil
- (d) Calcareous soil

**U.P.R.O./A.R.O. (Pre) 2014**

**Ans. (c)**

Alluvial soil is the most fertile soil in India. It has a lower amount of Nitrogen. Alluvial soils are formed mainly by Himalayan sediments and sea retreat. They cover 15 lakh sq. km of land.

**3. Which one of the following soils is deposited by rivers?**

- (a) Red soil
- (b) Black soil
- (c) Alluvial soil
- (d) Laterite soil
- (e) None of these/More than one of these

**63<sup>rd</sup> B.P.S.C. (Pre) 2017**

**Ans. (c)**

Alluvial soil is formed by the silt left by the rivers. Alluvial soils are divided into two sub-categories-

- (1) New alluvium or Khadar is formed in the floodplain of the river, where a new layer of soil is deposited every year during floods.
- (2) Ancient Alluvium or Bangar: The condition of the soil is at some height away from the reach of floods.

**4. Which of the following statements is/are true for alluvial soil?**

- I. Its sand content decreases from west to east.
- II. They are in the form of Bhangar and Khadar.
- III. This soil covers about 40 percent of the total area of the country.
- IV. This soil is rich in potash and poor in phosphorus.

Select the correct answer from the options given below:

- (a) I only
- (b) I and II only
- (c) I, II and III only
- (d) I, II, III and IV

**Jharkhand P.C.S. (Pre) 2023**

**Ans. (d)**

Alluvial soil is ideal for horticultural crops. It consists of material deposited by rivers- during floods. As we move inland towards the river valleys, soil particles appear somewhat bigger in size. According to their grains/components soils are described on the basis of their age. According to their age alluvial soils can be classified as old alluvial (Bangar and new alluvial (Khadar). Significantly, the sand content of this soil decreases from west to east. Notably, this soil covers about 40 percent (According to NCERT) of the total area of the country.

It is worth mentioning here that, mostly these soils contain an adequate proportion of potash, phosphoric acid and lime which are ideal for the growth of sugarcane, paddy, wheat and other cereal and pulse crops.

**5. In India, the largest soil group is found as :**

- (a) Red Soil
- (b) Black Soil
- (c) Sandy Soil
- (d) Alluvial Soil

**M.P.P.C.S. (Pre) 2006**

**Ans. (d)**